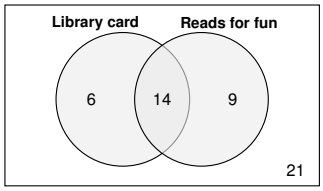
Unit 3 Review KEY

Name _____

One survey, every tool. 50 students answered two questions: *Do you have a library card? Do you read for fun?*

3.1 Read the table.
Each cell counts students who match **both** labels. Each row and each column adds to its total.

	Reads	Does not read	Total
Card	14	6	20
No card	9	21	30
Total	23	27	50



14 students have a card **and** read for fun.
Check: 14 + 6 = 20 and 14 + 9 = 23.
The Venn diagram shows the same counts. 21 students are outside both circles.

3.1 Divide by the grand total.
A relative frequency table shows the percent of **all** the students.

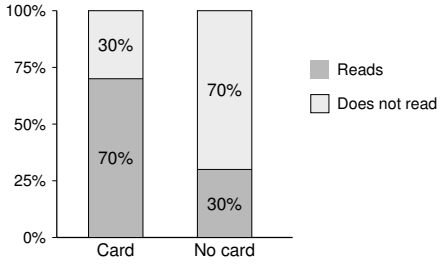
	Reads	Does not read	Total
Card	$\frac{14}{50}$ = 28%	$\frac{6}{50}$ = 12%	$\frac{20}{50}$ = 40%
No card	$\frac{9}{50}$ = 18%	$\frac{21}{50}$ = 42%	$\frac{30}{50}$ = 60%
Total	$\frac{23}{50}$ = 46%	$\frac{27}{50}$ = 54%	100%

28% of **all** the students have a card and read for fun. The whole table adds to 100%.

3.2 Marginal: use the totals.
Divide each total by the grand total.

Reads: $\frac{23}{50}$ = 46% Does not read: $\frac{27}{50}$ = 54%
Card: $\frac{20}{50}$ = 40% No card: $\frac{30}{50}$ = 60%

3.2 Conditional: use one group.
Divide by that group's total.
Different percents in the groups mean the two variables are **associated**.



Card: $\frac{14}{20}$ = 70% read for fun.
No card: $\frac{9}{30}$ = 30% read for fun.

70% and 30% are very different. Having a library card and reading for fun are **associated**.

Check the denominator. "Of the students with a card, what percent read for fun?" Divide by the card total (20). Do not divide by 50. Do not divide by the reads total (23).

An association does not prove that one thing causes the other.

Work with a partner. Compare answers when you finish each problem.

3.1 - Two-Way Tables and Relative Frequency

1. **I read a two-way table, and I fill in missing numbers in it.**

   120 students in 9th and 10th grade said which after-school club they are in. Each student is in one club.

   (a) Fill in the missing numbers.

	Art	**Chess**	**Garden**	**Total**
9th grade	18	22	15	55
10th grade	26	14	25	65
Total	44	36	40	120

   10th art: $44 - 18 = 26$. 10th total: $26 + 14 + 25 = 65$. 9th garden: $40 - 25 = 15$.
   9th chess: $55 - 18 - 15 = 22$. Chess total: $22 + 14 = 36$. Check: $55 + 65 = 120$.

   (b) How many students are in the chess club? 36

   (c) How many 9th graders are **not** in the art club? $55 - 18 = 37$

   (d) Which club has the most 10th graders? Art (26 students)

   75 students in 11th and 12th grade said what kind of phone they have.

   (e) Fill in the missing numbers.

	Android	**iPhone**	**Total**
11th grade	16	14	30
12th grade	20	25	45
Total	36	39	75

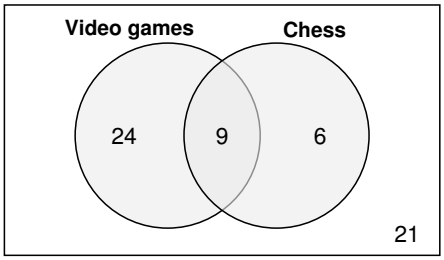
   11th iPhone: $30 - 16 = 14$. 12th iPhone: $39 - 14 = 25$. 12th total: $75 - 30 = 45$.
   12th Android: $45 - 25 = 20$. Android total: $16 + 20 = 36$.

   (f) How many 12th graders have an iPhone? 25

   (g) How many students have an Android phone? $16 + 20 = 36$

2. **I move between a Venn diagram and a two-way table, and I make a relative frequency table.**

This Venn diagram shows how many of 60 students play video games and how many play chess.



(a) Make a two-way table from the Venn diagram.

	Chess	No chess	Total
Games	9	24	33
No games	6	21	27
Total	15	45	60

(b) Make a relative frequency table. Divide each count by 60. Write percents.

	Chess	No chess	Total
Games	$\frac{9}{60} = 15\%$	$\frac{24}{60} = 40\%$	$\frac{33}{60} = 55\%$
No games	$\frac{6}{60} = 10\%$	$\frac{21}{60} = 35\%$	$\frac{27}{60} = 45\%$
Total	$\frac{15}{60} = 25\%$	$\frac{45}{60} = 75\%$	100%

(c) What percent of all the students play chess but not video games?

$\frac{6}{60} = 10\%$

(d) What percent of all the students play at least one of the two games?

$24 + 9 + 6 = 39$ students.
$\frac{39}{60} = 65\%$
(Or: $100\% - 35\% = 65\%$)

(e) Which statement is supported by the relative frequency table? Circle the letter.

A 9% of the students play both video games and chess.

(B) 25% of the students play chess.

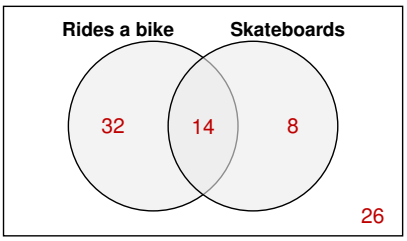
C 40% of the students play neither game.

D 10% of the students play video games but not chess.

Problem 2, continued.

80 students were asked if they ride a bike and if they skateboard. 46 ride a bike. 22 skateboard. 14 do both.

(f) Fill in the Venn diagram.



(g) Fill in the two-way table.

	Skate	**No skate**	**Total**
Bike	14	32	46
No bike	8	26	34
Total	22	58	80

(h) How many students do neither?

$80 - (32 + 14 + 8) = 26$

(i) What percent of all the students ride a bike and skateboard?

$\frac{14}{80} = 17.5\%$

3.2 - Marginal and Conditional Distributions

3. **I find a marginal distribution and a conditional distribution.**

120 students in 9th and 10th grade chose their favorite kind of game.

	Board	**Video**	**Card**	**Total**
9th grade	10	22	8	40
10th grade	26	34	20	80
Total	36	56	28	120

(a) Find the marginal distribution of favorite kind of game.

Board $\frac{36}{120} = 30\%$, Video $\frac{56}{120} \approx 47\%$, Card $\frac{28}{120} \approx 23\%$

(b) Find the marginal distribution of grade.

9th grade $\frac{40}{120} \approx 33\%$, 10th grade $\frac{80}{120} \approx 67\%$

(c) Find the conditional distribution of favorite game for 9th graders.

Board $\frac{10}{40} = 25\%$, Video $\frac{22}{40} = 55\%$, Card $\frac{8}{40} = 20\%$

(d) Find the conditional distribution of favorite game for 10th graders.

Board $\frac{26}{80} = 32.5\%$, Video $\frac{34}{80} = 42.5\%$, Card $\frac{20}{80} = 25\%$

(e) Find the conditional distribution of grade for the students who chose card games.

Divide by the card total, 28. 9th grade $\frac{8}{28} \approx 29\%$, 10th grade $\frac{20}{28} \approx 71\%$

Is each statement a **marginal** or a **conditional** percent? Circle one.

(f) About 47% of all the students chose video games. (Marginal) Conditional

(g) Of the 10th graders, 25% chose card games. Marginal (Conditional)

(h) 30% of the students chose board games. (Marginal) Conditional

(i) 55% of 9th graders chose video games. Marginal (Conditional)

4. **I choose the right total to divide by, and I decide if two categorical variables are associated.**

A class grew 60 bean plants. Some plants got plant food. The class recorded which plants grew flowers.

	Flowers	**No flowers**	**Total**
Plant food	30	10	40
No plant food	9	11	20
Total	39	21	60

(a) What percent of the plants with plant food grew flowers? $\frac{30}{40} = 75\%$

(b) What percent of the plants without plant food grew flowers? $\frac{9}{20} = 45\%$

(c) Is plant food associated with growing flowers? Explain.
Yes. The percents are different. 75% of the plants with plant food grew flowers, but only 45% of the plants without it.

(d) What percent of the plants with flowers got plant food? $\frac{30}{39} \approx 77\%$

(e) Why are the answers to (a) and (d) different?
They divide by different groups: (a) by the 40 plants with plant food, (d) by the 39 plants with flowers.

(f) Of the plants with no flowers, what percent got no plant food? Which fraction gives the answer? Circle the letter.

A $\frac{11}{20}$ (B) $\frac{11}{21}$ C $\frac{11}{60}$ D $\frac{21}{60}$

Problem 4, continued.

130 students in 9th and 10th grade said if they write with a pen or a pencil.

	Pen	Pencil	Total
9th grade	15	35	50
10th grade	24	56	80
Total	39	91	130

(g) What percent of 9th graders use a pen? $\frac{15}{50} = 30\%$

(h) What percent of 10th graders use a pen? $\frac{24}{80} = 30\%$

(i) Is grade associated with using a pen or a pencil? Explain.
No. The percents are the same. 30% of the 9th graders and 30% of the 10th graders use a pen.

Test-Style Questions

A test asks the same questions with more words. Read slowly. Circle the best answer.

5. A survey asked 90 students in 11th and 12th grade where they like to watch movies: at a theater or at home. Each student chose one place. The two-way table shows the results, but some of the values are missing.

	Theater	Home	Total
11th grade	12		40
12th grade			
Total	30		90

Based on the table, how many 12th graders like to watch movies at home?

A 28
(B) 32
C 50
D 60

12th grade total: $90 - 40 = 50$. 12th grade, theater: $30 - 12 = 18$. 12th grade, home: $50 - 18 = 32$.

6. A school surveyed 200 students about their phones. The two-way table shows whether each phone has a case and whether its screen is cracked.

	Cracked	Not cracked	Total
Case	14	106	120
No case	36	44	80
Total	50	150	200

What percent of all the students surveyed have a phone with a case and a cracked screen?

(A) 7%

B about 12%

C 25%

D 28%

"All the students" means divide by the grand total: $\frac{14}{200} = 7\%$.

7. A music teacher asked 150 students in 11th and 12th grade which instrument they would most like to learn. The results are shown in the two-way table.

	Guitar	Piano	Drums	Total
11th	30	18	12	60
12th	36	45	9	90
Total	66	63	21	150

Based on the table, what percent of the 12th graders surveyed want to learn piano?

A 30%

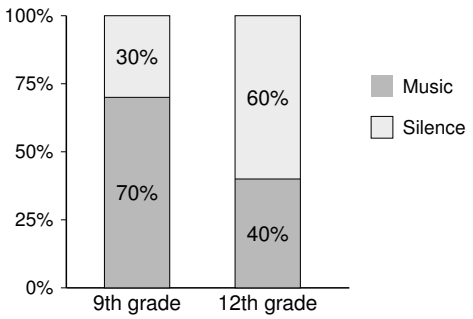
B 42%

(C) 50%

D about 71%

Divide by the 12th grade total: $\frac{45}{90} = 50\%$.

8. A survey asked students in 9th grade and 12th grade if they like to study with music or in silence. The segmented bar chart shows the conditional distribution of study choice for each grade.



Which statement is supported by the segmented bar chart?

A There is no association, because each bar adds up to 100%.

(B) Grade and study choice are associated, because the percent of students who study with music is different in the two grades.

C More 9th graders than 12th graders were surveyed.

D Most 12th graders like to study with music.